

7th Semester of B. Pharm. Examination
University Theory Examination November/December 2017
PH420.6 Sterile Product Technology-II

Date: 01.12.2017, Friday Time: 10:00 a.m. To 12:00 Noon Maximum Marks: 50

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1	Answer the following:	05
A	Write any two major advantages of In-process quality control tests for parenteral products.	01
B	Define: Parenteral formulation.	01
C	Suggest the suitable material for packaging of ophthalmic drops.	01
D	How the 'Risk assessment' for manufacturing of a sterile formulation is beneficial?	01
E	Justify: Type-IV glass can not be used for the manufacturing process and packaging of parenteral formulation.	01
Q 2	Attempt any <u>TWO</u> of the following	
A	Write a brief note on: Blow Fill Seal Technology as a method for manufacturing and packaging of parenteral products.	05
B	Explain in detail the manufacturing method and applications of following sterile dosage forms with examples:	05
	(a) Emulsion	
	(b) Dry Powders for reconstitution	
C	Compare and Contrast: Small Volume Parenterals and Large Volume Parenterals.	05
Q 3	Attempt any <u>TWO</u> of the following	
A	Enlist the formulation components and manufacturing equipments with their applications for the following sterile dosage forms:	05
	(a) Contact lens solution	
	(b) Eye lotion	

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| B | Explain the manufacturing method and packaging process of small volume Parenterals. | 05 |
| C | (i) Write about the importance of packaging integrity for sterile products.
(ii) Explain the product development steps for formulation of ophthalmic Ointment as per ICHQ8 guideline. | 05 |

SECTION- II

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| Q 4 | Answer the following: | 05 |
| A | Comment: LAL test is more preferable than rabbit test for evaluation of sterile dosage forms. | 01 |
| B | Comment: I.V. is the most popular route of parenteral administration. | 01 |
| C | Comment: Chelating agent should be used in ophthalmic dosage forms. | 01 |
| D | Enumerate any two examples of antimicrobial preservatives used in ophthalmic dosage form. | 01 |
| E | Which culture medias are used for sterility test of sterile dosage forms as per I.P? | 01 |
| Q 5 | Attempt any <u>TWO</u> of the following | |
| A | Enlist evaluation parameters for injectable dosage form and explain any one in brief. | 05 |
| B | (i) Explain the preparation process and storage of Water for Injection.
(ii) Explain the disadvantages of injectable dosage forms. | 03
02 |
| C | Discuss in brief different routes of parenteral administration. | 05 |
| Q 6 | Attempt any <u>TWO</u> of the following | |
| A | Classify the different types of glass with their applications and explain any one evaluation test of it. | 05 |
| B | Write a note on formulation of ophthalmic drops as sterile dosage form. | 05 |
| C | Discuss different approaches for parenteral drug delivery. | 05 |